Manual Check

1) Grid fits both plates

$\frac{30}{5} = 6 \quad \frac{10}{5}$ ✓

2) $q0 \cdot \Delta x / k_1$

$750 \cdot 0.005 / 36 = 0.10416$ ✓

3) $q0 \cdot \Delta x / k_2$

$750 \cdot 0.005 / 13 = 0.28846153$ ✓

4) $h \Delta x / k_2$ original $= 0.002984\overline{6}$ ✓

$7.76 \times 0.005 / 13$

5) Interface coefficients

$36, -(36+13), 13 = 36, -49, 13$ ✓

6) T8 original

$20 + 750 / 7.76 = 116.649$ ✓

7) T6 original

$T8 + 750 \times 0.01 / 13 = 117.226$ ✓

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8) TO original

$T_6 + 750 \times 0.03136 = 117.951$ ✓

9) Interface flux continuity

$36 \times 0.104167 / 0.005$; $B \times 0.288462 / 0.005$

$= 750.0024$ ✓ ; $= 750.0012$ ✓

10) Energy Balance

$7.76 (116.649 - 20) = 749.996$ ✓

11) $q_{O,max}$, original

$731 / 1.304685 = 559.522$ ✓

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Audit 12-22

12) Minimum h_comb

$$\frac{1}{\left(\frac{73}{750} - 0.0016026\right)} = 10.445966$$

ok

13) T_8 and T_{10} adapted

$$\frac{20+750}{40} = 38.750 + 0.577 + 0.625$$

ok

$$= 39.952 \text{ ok}$$

14) q_0, max, adapted

$$\frac{73}{(0.025 + 0.0016026)} = 2744.09268 \text{ ok}$$

15) Biot numbers

$$7.76(0.0016026) = 0.12436176 \text{ ok}$$

$$40(0.0016026) = 0.064104 \text{ ok}$$

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16) Wall heat capacity C''

$(8530)(377)(0.03) + (8000)(500)(0.01$

$= 136474.3$ ok

17) Time constants C''/h

$\frac{136474}{7.76} = 17586.85567$ ok

$\frac{136474}{40} = 3411.85$ ok

18) Mesh Fourier number τ_1

$\left(\frac{36}{3215810}\right)\left(\frac{1}{(0.005)^2}\right)$

$= 0.4477876491$ ok

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19) First explicit step T_8 at 1 s

$$93 - 0.26 \times 0.0029846 \times 73$$
$$= 92.94335229 \quad \text{ok}$$

20) One-term root, original

$$0.11129 \times \tan(0.11129)$$
$$= 0.01243685 \quad \text{ok}$$

21) One-term validity

$Fo = 104.7 ; 20.8$
requires $Fo > 0.2$
both ok

22) Cooldown improvement

$$\frac{(22,927.7 - 4575.0)}{22,927.7}$$

$$= 0.80045970 \quad \text{ok}$$

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